EXPERIMENT – 7

AIM

To develop a program that demonstrates data-link-layer flow control using the Sliding Window Protocol and models the transfer of frames between two nodes.

THEORY

The Sliding Window Protocol is a flow-control method used at the data link layer. It enables a sender to transmit a group of frames before it needs to wait for acknowledgements from the receiver.

For this experiment, every character in the entered message is treated as an individual frame. Each frame has a frame number and a data character. The number of frames sent at a time depends on the window size entered by the user.

The receiver obtains the frames from **Sender_Buffer** and verifies that their frame numbers arrive in the expected sequence. A valid frame receives an ACK, while an unexpected frame causes a NACK.

The sender checks the acknowledgements saved in **Receiver_Buffer**. When the expected ACKs are received, the window advances and the next group of frames is sent. If a NACK appears, the required frames are sent again.

The experiment can also be used to demonstrate errors by manually editing a frame number or acknowledgement number in the relevant buffer file, as described in the lab instructions.

ALGORITHM

Sender

1. Begin execution.
2. Get the window size from the user.
3. Read the message to be transmitted.
4. Split the message so that each character forms one frame.
5. Give each frame a unique number.
6. Build each frame using its number and character data.
7. Transmit frames in groups based on the window size.
8. Write the sent frames to Sender_Buffer.
9. Wait for acknowledgement information from the receiver.
10. Open and read Receiver_Buffer.
11. Verify the acknowledgement values.
12. If the expected ACKs are present, shift the window forward.
13. Transmit the next group of frames.
14. If a NACK is found, retransmit the necessary frame or frames.
15. Repeat until every frame has been transmitted successfully.
16. End the program.

Receiver

1. Start the receiver process.
2. Read the transmitted frames from Sender_Buffer.
3. Inspect the number attached to each frame.

4. Compare each number with the next expected frame number.

5. If the number is correct, accept the frame and create an ACK.

6. If the number is incorrect, create a NACK.

7. Save the acknowledgement in Receiver_Buffer.

8. Continue checking frames until the complete message is received.

9. Stop the receiver process.

PROGRAM

```
import time

message = input("Enter the message: ")
window = int(input("Enter window size: "))

frames = []
for i, ch in enumerate(message):
    frames.append([i + 1, ch])

i = 0
while i < len(frames):
    end = min(i + window, len(frames))

    print("\nSending frames:")
    with open("Sender_Buffer", "w") as f:
        for j in range(i, end):
            print("Frame No:", frames[j][0], "Data:", frames[j][1])
            f.write(str(frames[j][0]) + " " + frames[j][1] + "\n")

    time.sleep(1)
    expected = i + 1

    with open("Receiver_Buffer", "w") as f:
        for j in range(i, end):
            if frames[j][0] == expected:
                f.write("ACK " + str(frames[j][0]) + "\n")
                print("Receiver: ACK", frames[j][0])
                expected += 1
            else:
                f.write("NACK " + str(expected) + "\n")
                print("Receiver: NACK", expected)
                break

    with open("Receiver_Buffer", "r") as f:
        ack = f.readlines()

    success = True
    for line in ack:
        parts = line.split()
        if parts[0] == "NACK":
            success = False
            break

    if success:
        i = end
    else:
        print("Retransmitting frames...")
        time.sleep(1)

print("\nAll frames transmitted successfully.")
print("Message:", message)
```

SAMPLE INPUT

```
Enter the message: HELLO
Enter window size: 3
```

SAMPLE OUTPUT

```
Sending frames:

Frame No: 1 Data: H
Frame No: 2 Data: E
Frame No: 3 Data: L
```

```
Receiver: ACK 1
Receiver: ACK 2
Receiver: ACK 3

Sending frames:

Frame No: 4 Data: L
Frame No: 5 Data: O

Receiver: ACK 4
Receiver: ACK 5

All frames transmitted successfully.
Message: HELLO
```

OBSERVATION

Frame No.	Data	Acknowledgement
1	H	ACK 1
2	E	ACK 2
3	L	ACK 3
4	L	ACK 4
5	O	ACK 5

With a window size of 3, the sender first transmits three frames. After the receiver acknowledges them, the window advances and the remaining frames are sent.

When a frame number is altered or received out of sequence, the receiver issues a NACK, and the sender repeats the required transmission.

RESULT

Therefore, the Sliding Window Protocol was implemented successfully, and frame transfer between the sender and receiver was demonstrated using acknowledgements and retransmission.